# ThermIQ

Industrial Knowledge Intelligence for Thermal Power Plants

*A unified asset & operations brain that makes a plant's scattered documents queryable —*
***and quantifies undocumented knowledge as financial risk.***

**ET AI Hackathon 2026 · Problem 8**
AI for Industrial Knowledge Intelligence — Unified Asset & Operations Brain

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PROBLEM CONTEXT

# The Industrial Knowledge Gap

Asset-intensive plants run on knowledge scattered across P&IDs, maintenance logs, operating procedures, inspection reports and regulatory filings — typically 7–12 disconnected systems per plant. Much of the critical operating judgement is never written down at all; it lives in the heads of senior engineers.

The cost is threefold: hours lost re-finding or recreating information, unplanned downtime from decisions made without full equipment history, and a looming knowledge cliff as experienced staff retire and take undocumented expertise with them.

***This is not a document-management problem.*** *It is a safety, quality and operational-efficiency problem — and it compounds every year it is left unsolved, across India's 200+ GW coal fleet.*

**35%**

of working hours spent searching for or recreating documents that already exist

*McKinsey, 2024*

**7–12**

disconnected document systems in the average large plant

*NASSCOM–EY*

**18–22%**

of unplanned downtime events linked to knowledge fragmentation

*BIS Research*

**~25%**

of experienced engineers retiring within the next decade

*Sector estimate*

SOLUTION OVERVIEW

# A Unified Asset & Operations Brain

ThermIQ ingests a plant's heterogeneous documents — text and scanned PDFs (via OCR), spreadsheets, and Google Drive folders — into one searchable knowledge base, then layers four capabilities on top. Its defining move: it does not only retrieve what is documented, **it detects and prices what is missing.**

**1**

### Query Copilot

Ask questions in plain English; get direct, cited answers drawn from the plant's own documents and the benchmark standards.

**2**

### Knowledge-Gap Detection

Scores each plant's corpus against a benchmark of what should be documented, flagging what is absent or only partial.

**3**

### Risk Quantification Dashboard

Converts every gap into rupee-crore operational risk, ranked, with a suggested closure order and a one-click Excel export.

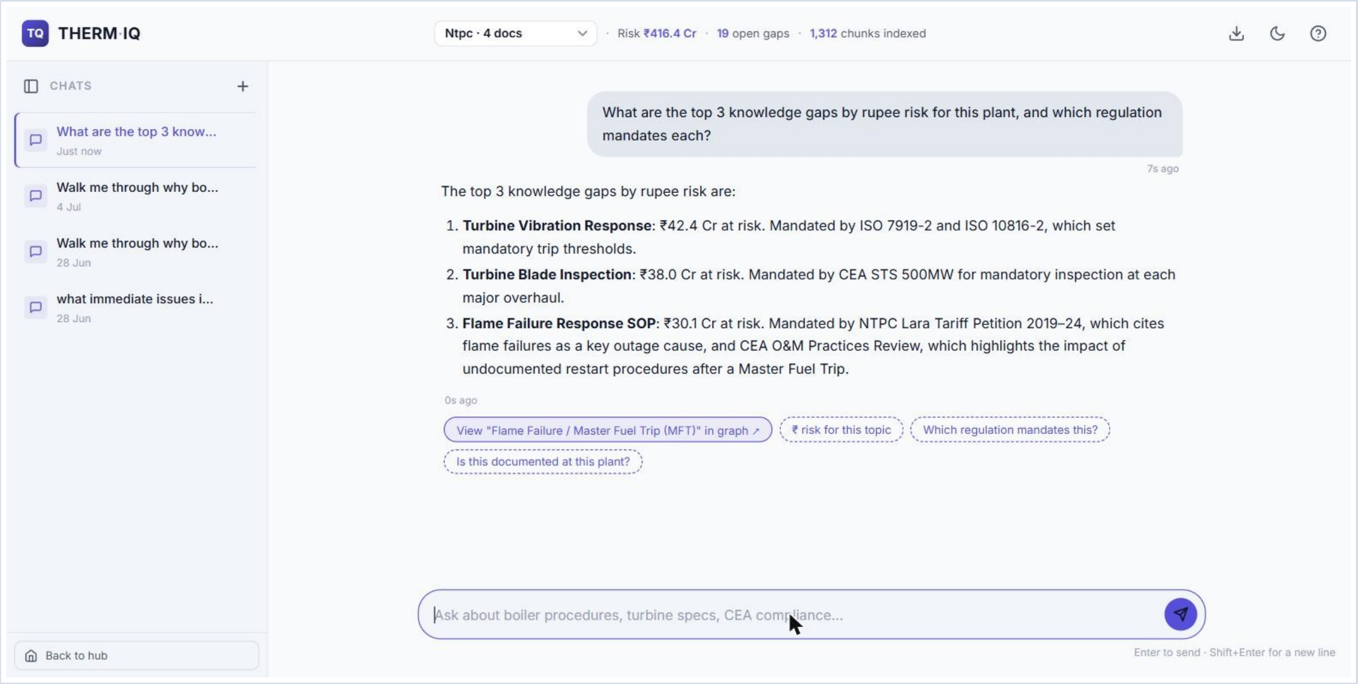
**4**

### Knowledge Graph

Links failure modes to real historical outage cost and the regulations that mandate the missing procedure.

***One ingestion pipeline, one live backend, one risk model.*** *The four modules are views on the same data — not four separate demos.*

# Conversational Knowledge Retrieval




**UNDER THE HOOD** Jina AI embeddings · Qdrant vector search · Gemini 2.5 Flash · agentic tool-calling

*Example — an NTPC plant profile: the assistant returns the top gaps with ₹ exposure and the regulation mandating each.*

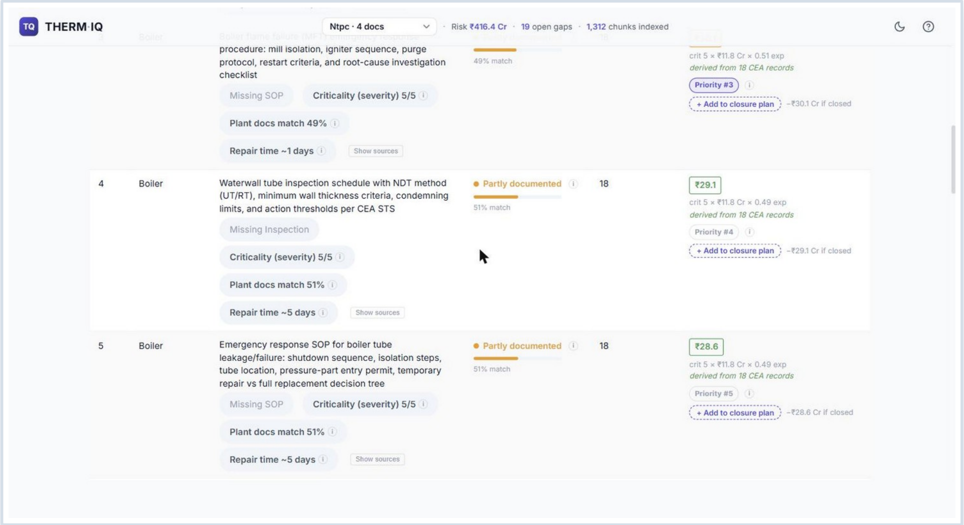
- **Ask in plain English.** Any procedure, spec or failure mode — a direct, cited answer in seconds, no keyword hunting.
- **Understands meaning, not words.** Semantic search matches intent even when the wording differs (Jina AI + Qdrant).
- **Answers you can trust.** Every response separates benchmark standards from the plant's own documents, with source citations and a confidence signal.
- **Decides what to look up.** Agentic tool-use picks between document search, the live risk register, real outage records, or the web.
- **Always available.** A four-model fallback chain (Gemini → NVIDIA NIM → OpenRouter) keeps it answering under free-tier limits.
- **Guides the next step.** Multi-turn memory, one-click follow-up suggestions, and a jump straight into the knowledge graph.

FEATURE DEEP-DIVE · 2

# Quantifying Knowledge Gaps as Financial Risk

**Risk** = **Criticality** (how severe, 1–5) × **Consequence** (₹ Cr lost per outage) × **Exposure** (how undocumented, 0–1)

| | |
|---|---|
| **Criticality** | Graded from CEA forced-outage frequency and CERC regulations — evidence, not opinion. (e.g. boiler failures lead forced outages; Vasudha 2022, 735 units.) |
| **Consequence** | The real revenue lost when this fails, computed from CEA daily outage records at ₹5/kWh (LBNL/Ember, 2024). |
| **Exposure** | How poorly the plant documents this topic — the gap between the question and the plant's own documents (Jina AI similarity). |

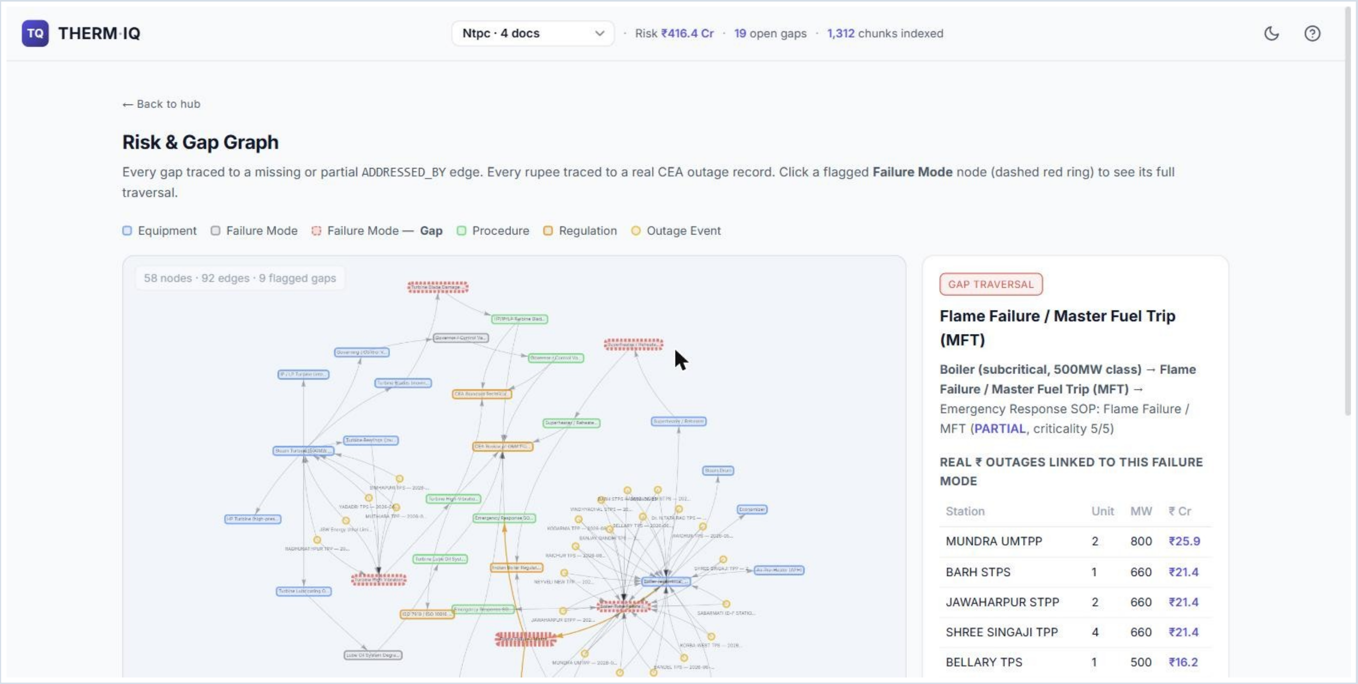



- **19-item expert checklist.** What a well-run plant should document — scored automatically against each plant's corpus.
- **Ranked by rupee impact.** Every gap ordered by ₹ exposure, with coverage status and a recommended closure sequence.
- **Board-ready output.** One-click themed Excel export of the full risk register, per plant.

**Worked example (NTPC profile).** The engine surfaces 19 gaps totalling ₹416.4 Cr of exposure — every figure traceable to its CEA / CERC source. The same engine runs unchanged on any plant's documents.

FEATURE DEEP-DIVE · 3

# Failure-Mode-to-Regulation Traceability




**UNDER THE HOOD** Neo4j knowledge graph · equipment → failure mode → outage event → regulation

- **A map, not a list.** A knowledge graph (Neo4j) connects equipment, failure modes, real outage events and the regulations that govern them.
- **Click to trace.** Select any flagged failure mode to see how severe it is, how much it has already cost across the fleet, and which rule mandates documenting it.
- **Turns a number into evidence.** A single ₹ figure becomes an auditable chain a plant or auditor can defend.

EXAMPLE TRACE · NTPC

**Waterwall tube thinning** flagged PARTIAL · criticality 5/5

**18 real CEA outage records** = ₹211.85 Cr already lost across the fleet

**Mandated by** CEA STS 500 MW + Indian Boiler Regulations

# Technology & Data Flow

## Frontend

Single-page web app (GitHub Pages + Vercel); works on desktop and field mobile.

## Backend

Lightweight serverless functions on Vercel.

## Understanding docs

Jina AI turns text into vectors; Qdrant searches them by meaning.

## Answers

Google Gemini 2.5 Flash, with NVIDIA NIM + OpenRouter as automatic backups.

## Records & structure

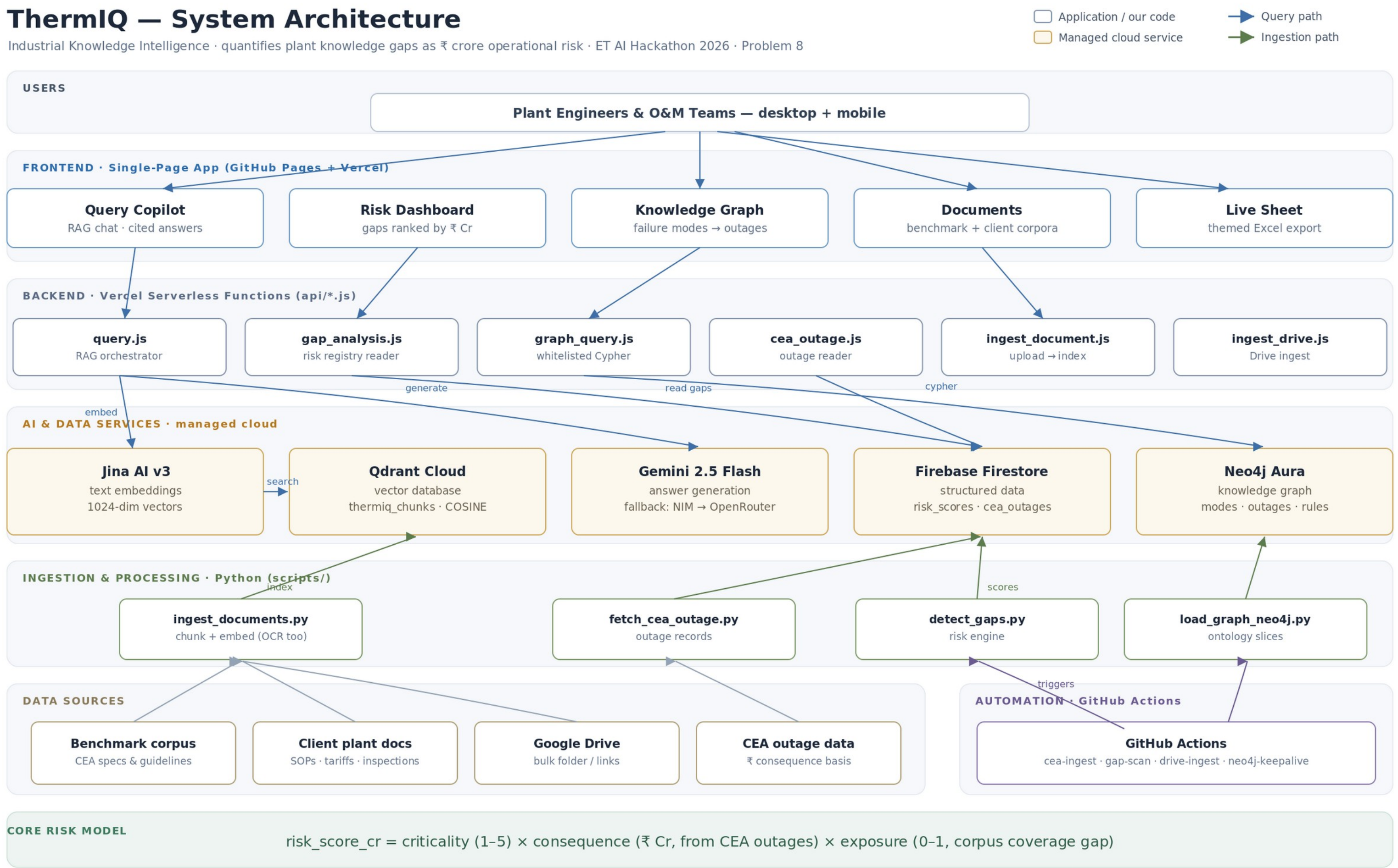
Firebase Firestore (risk register, outage data) · Neo4j (knowledge graph).

## Ingestion

Python pipelines chunk & embed PDFs, run OCR on scans, and score gaps.

## Automation

GitHub Actions refresh CEA outage data and re-scan gaps on a schedule.




*Public CEA standards & outage reports form the fixed benchmark; each plant's own uploaded documents form the client corpus measured against it.*

DIFFERENTIATION

# What Sets ThermIQ Apart

### Prices the absence of knowledge

Most tools retrieve what is written down. ThermIQ scores what is missing and puts a rupee value on it — reframing documentation from a compliance chore into a capital-allocation decision.

### Grounded in real data, end to end

Every rupee figure traces to actual CEA outage records and CERC regulation — no invented benchmarks or fabricated accuracy claims.

### Benchmark-vs-client architecture

Public CEA standards are the fixed yardstick; each plant's own SOPs are measured against it — so onboarding a new plant needs zero re-engineering.

### Multi-plant by design

Each plant is an isolated profile with its own corpus and scores (namespacing) — demonstrated across separate NTPC and Saraighat profiles.

### Resilient by construction

A four-model answer-generation fallback chain keeps the copilot live under free-tier limits; the graph auto-recovers via a scheduled keep-alive.

### Auditable by default

Every score and answer carries its source, so a plant manager can defend it — for instance, in a CERC tariff petition.

DELIVERY STATUS

# Live Today · Limitations · Roadmap

## ● Live today

- Single-page app with five working views
- RAG copilot with source citations
- 19-item gap-scoring engine (v3)
- Rupee risk register, ranked & exportable
- Neo4j graph with click-through traversal
- Per-plant profiles (NTPC, Saraighat)
- Automated CEA-ingest & gap-scan pipelines

## ▲ Known limitations

- One source spec (BMD-01) unreachable from the build network
- No computer-vision P&ID parsing yet — text + OCR only
- Free-tier hosting caps backend functions
- A finer v4 scoring engine is built but held back pending human review of two outliers
- Accuracy not yet scored against a formal expert benchmark

## → Roadmap

- Computer-vision parsing of P&IDs and engineering drawings
- More plants onboarded at full document depth
- Mobile-first field-technician view
- Validated v4 evidence-graded scoring engine
- Expert-benchmarked answer & extraction accuracy

**Shipped and honest:** v3 is the validated scoring engine running live today. The finer v4 engine and computer-vision drawing parsing are the immediate next milestones — deliberately not overstated as done.

CONCLUSION

# An intelligence layer for industrial knowledge — buildable today

*ThermIQ answers Problem 8 directly: it ingests heterogeneous industrial documents, makes their collective knowledge queryable with citations, and — uniquely — quantifies what remains undocumented as financial risk, every figure traceable to real regulatory and outage data.*

*The result: a plant can see, in rupees, what its knowledge gaps are worth, and close the highest-value ones first.*

### Retrieve

Cited answers across the full document corpus, for engineers and O&M teams.

### Quantify

Every knowledge gap priced as ₹-crore operational risk, ranked and sourced.

### Trace

Failure mode → real outage cost → governing regulation, end to end.

Live application **therm-iq.vercel.app** Source **github.com/GhostUnamused/thermIQ**

*Yaminichandra K J · IIM Amritsar (IPM) · a research-preview prototype built with AI as a development partner*